$$\bar{X} = A + \left[\frac{\sum fd}{\sum f} \times c\right], \text{ where } d = \frac{x - A}{c}$$

Example 8.3 Find the mean of the following distribution using Step Deviation Method.

Class Interval	0-8	8-16	16-24	24-32	32-40	40-48
Frequency (f)	10	20	14	16	18	22

Solution

Let Assumed mean $A = 28$, class width c =8

Class Interval	Mid Value x	Frequency f	$d = \frac{x - A}{c}$	fd
0-8	4	10	–3	–30
8-16	12	20	–2	–40
16-24	20	14	–1	–14
24-32	28	16	0	0
32-40	36	18	1	18
40-48	44	22	2	44
		$\sum f = 100$		$\sum fd = -22$

Mean

$$\bar{X} = A + \frac{\sum fd}{\sum f} \times c$$

$$= 28 + \left(\frac{-22}{100}\right) \times 8$$

$$= 28 - 1.76 = 26.24$$

Note

- When x_i and f_i are small, then Direct Method is the appropriate choice.
- When x_i and f_i are numerically large numbers, then Assumed Mean Method or Step Deviation Method can be used.
- When class sizes are unequal and d numerically large,we can still use Step Deviation Method.

8.4.4 A special property of the Arithmetic Mean

1. *The sum of the deviations of the entries from the arithmetic mean is always zero.*

If $x_1, x_2, x_3, \ldots, x_n$ are n observations taken from the arithmetic mean $\bar{X}$

then $(x_1 - \bar{X}) + (x_2 - \bar{X}) + (x_3 - \bar{X}) + \ldots + (x_n - \bar{X}) = 0$ .Hence $\sum_{i=1}^{n} (x_i - \bar{X}) = 0$

2. *If each observation is increased or decreased by k (constant) then the arithmetic mean is also increased or decreased by k respectively.*

3. *If each observation is multiplied or divided by k, $k \neq 0$, then the arithmetic mean is also multiplied or divided by the same quantity k respectively.*

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